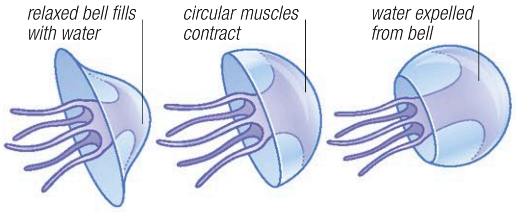
In this kind of skeleton, some bones meet at flexible joints. Other bones, particularly those in the skull, lock together for extra strength.

Muscles and movement

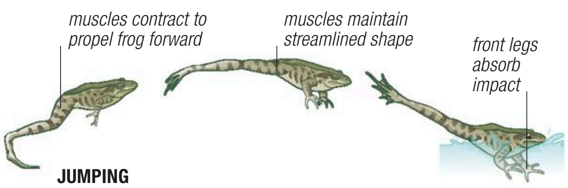
Muscles work by contracting. This means they can pull but not push. In most cases, they are arranged in pairs or groups that pull in opposing directions: when one muscle or muscle group contracts, its partner is brought back to its normal resting shape.

Muscles make animals move in different ways. In animals without limbs, such as earthworms and jellyfish, they work to change the body's shape. In earthworms, opposing muscles alternately shorten and lengthen the animal's segments so that it can creep through soil. In limbed animals, one set of muscles pulls the limb down or back, while the other lifts it up or forward.

As well as making animals move, muscles serve other purposes. They force food through the digestive system (peristalsis) and pump blood around the circulatory system. Unlike most other muscles, the heart muscle has a built-in rhythm that keeps it contracting throughout an animal's life.



MOVING WITHOUT LIMBS
Jellyfish swim by rhythmically contracting the bell-shaped part of their body. This expels water from the bell, which pushes the jellyfish forward. Most jellyfish make little headway against the current: they swim mainly to keep at the right level in the water.



JUMPING
When a frog leaps, its legs act as levers, propelling it into the air. Its front legs fold up to cushion its body from the impact when it lands.

Body coverings

Animal cells are easily damaged. To protect them from injury and disease, animals have body coverings, most of which consist largely of nonliving matter. Mammalian skin is covered by dead cells, while insect body cases are covered by a hard substance called "chitlin" and waterproof wax. In many cases, these protective layers are themselves protected: mammals often have a coat of fur, while many other animals have scales. Some of these extra coverings have developed additional uses. Soft feathers and fur help to retain body heat, while extra-strong feathers are used in flight. Colors or patterns act as camouflage or help animals recognize their own kind.


FEATHERS


SCALES


FUR

BODY BARRIERS
Bird feathers, butterfly scales, and mammal fur are made of nonliving substances that are produced by living cells. Feathers and fur are replaced during their owner's lifetime, but butterfly scales are not.

Respiration

For small and thin animals, obtaining oxygen is a simple matter because it seeps into their bodies from outside. At the same time, carbon dioxide escapes in the other direction. For larger animals, respiration is more complex. In relative terms, they have a much greater volume than surface area, so there is less room for gases to move in and out. To breathe, they rely on respiratory organs—structures that effectively pack a large surface area into a small amount of space. In aquatic animals, gills are the most common respiratory organs. Typical gills consist of thin, flat, or feathery surfaces that bring blood into close contact with the water outside.

However, most gills do not work in air because—out of water—their surfaces collapse and stick together. Land animals therefore have hollow respiratory organs that carry air deep inside their bodies. In insects, these organs are tubes, called tracheae, which divide into extremely fine filaments that reach individual cells. In land-dwelling vertebrates, the organs are lungs—air-filled chambers surrounded by a network of blood vessels. Muscles make the lungs expand or contract, sucking air in or blowing it out.

EFFICIENT RESPIRATION
Animals that are highly active in cold habitats need lots of oxygen. When an ibex runs uphill, its lungs take in 10 times as much air as when it is lying down.

Nerves and senses

Nerve cells, or neurons, are the animal world's equivalent of wiring. Neurons conduct brief bursts of electricity, known as impulses, which carry information from sense organs or make muscles contract. Corals and other simple animals have a network of nerves scattered throughout their bodies. But in most animals, the nervous system converges on the brain.

Some animal senses, such as touch, operate through nerve endings scattered all over the body. A similar sense, which works internally, tells animals about their posture. The most important senses—vision, smell, and hearing—work through organs that form some of the most elaborate structures in the body.

Vision is essential for many animals, and eyes show a wide variety of designs. At their simplest—for example, in snails—they do little more than distinguish between light and dark. In many animals, particularly arthropods and vertebrates, they focus light onto large numbers of nerve cells, building up a detailed image of the surroundings. In vertebrates, these eyes have a single lens, which throws light onto a "screen," or retina. In arthropods, the eye has up to 25,000 separate compartments, each with its own lens system; these compound eyes create a mosaic-like image and are especially good at detecting movement.

Mammals are the only animals with prominent earflaps. Vertebrates' ears are always on the head, but in some animals, they are positioned elsewhere. Most grasshoppers and crickets have ears on their abdomen or legs. Organs that detect taste and smell can also be in a variety of positions. Like ears, they can be used in communication, as well as for avoiding danger and finding food.

Many animals have senses that are more acute than those of human beings, and some can sense things that humans cannot. For example, most fish can sense pressure waves in water, and many can detect weak electric fields. Some snakes can "see" warmth, enabling them to attack warm-blooded prey in total darkness.

BINOCULAR VISION
Jumping spiders have 2 extra-large eyes that face directly forward. This gives them binocular vision, which is essential for gauging distances before making a jump.